

Computer Architectures

Exam of 25.2.2026 - part I

Question #1A

Let consider *dependences* in programs, which are known to be possible sources of *hazards*.

You are invited to

1. List the possible types of dependencies
2. For each type of dependency, report a fragment of RISC-V code where the dependency appears
3. For each type of dependency, explain in which case(s) that dependency may cause a hazard.

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Question #2A

Let consider the Branch Prediction mechanism based on the Branch Target Buffer (BTB).

You are requested to

1. Describe the architecture of a BTB, detailing the content of the memory it is based on
2. Describe the behavior of a BTB: when it is accessed, which input information items receives, and which output data it produces each time
3. Assuming that the processor uses 32 bit addresses, each instruction is 4 byte wide, and the BTB is composed of 16 entries, provide a table showing the final content of each entry of the BTB and compute the prediction done for each of the branch instructions if
 - The BTB is initially empty (i.e., full of 0s)
 - The following instructions are executed in sequence
 - add t1, t2, t3 located at the address 0x00002000
 - beq t2, t5, l1 located at the address 0x00002004; the branch is taken, and the branch target address is 0x00040024
 - addi t1, t0, #1 located at the address 0x00040024
 - bne t3, t4, l2 located at the address 0x00040028; the branch is taken, and the branch target address is 0x000AA004
 - sll t1, t0, #2 located at the address 0x000AA004
 - sub t2, t0, t5 located at the address 0x000AA008
 - beq t2, t3, l3 located at the address 0x000AA00C; the branch is not taken
 - beq t2, t5, l4 located at the address 0x000AA010; the branch is taken, and the branch target address is 0x00002000
 - add t1, t2, t3 located at the address 0x00002000
 - beq t2, t5, l1 located at the address 0x00002004; the branch is taken, and the branch target address is 0x00040024.

Index	Final BTB Content
0	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
14	
15	